

Department of Mathematics, IIT Madras MA-5105-Scientific Computing & Advanced Programming Lab Practice Problems 5 September 28, 2026 <i>Lecturer: Saurav Samantaray</i> <i>TA: Rohit Kumar Mani</i>

Instructions

You have already implemented the following basic numerical linear algebra algorithms discussed in the course:

- Gaussian elimination and pivoting,
- LU factorization,
- conditioning and error analysis,
- power iteration and the Rayleigh quotient,
- QR decomposition,
- QR iteration.

For the problems below, the numerical method is deliberately not prescribed. You should decide which algorithms are appropriate, implement the required computational experiment, and justify your choices.

Use Python and NumPy. Built-in routines may be used for verification, but the main computation should use algorithms developed in the course.

Problem 1: Recovering a Hidden Model from Noisy Data

An experiment measures the response of a system at the points

$$x_i = 0, 0.1, 0.2, \dots, 1.$$

The measured data are assumed to have the form

$$y_i = 2.0 + 1.5x_i - 0.8x_i^2 + 0.6x_i^3 + \eta_i,$$

where η_i represents measurement noise.

The data are supplied in the file `data.txt`.

You want to construct a polynomial approximation

$$p(x) = c_0 + c_1x + c_2x^2 + c_3x^3.$$

Tasks

1. Formulate the problem as a numerical linear algebra problem.
2. Design a computational procedure to determine

$$c_0, c_1, c_2, c_3.$$

3. Choose an appropriate numerical method from those studied in the course. Explain briefly why you selected it.
4. Repeat the computation using a different numerical method and compare the results.
5. For each computed polynomial, evaluate the residual

$$r = y - Xc,$$

and report an appropriate norm of the residual.

6. Now increase the degree of the polynomial. Repeat the experiment for degrees

$$5 \quad \text{and} \quad 8.$$

7. Investigate the conditioning of the resulting linear algebra problem as the polynomial degree increases.
8. Compare the computed coefficients, residuals, and numerical behaviour for the different polynomial degrees.
9. Give a short explanation of what your numerical experiments reveal about fitting data using polynomial models.

Question to think about

A smaller residual does not necessarily mean that the computed model is numerically more reliable. Use your experiments to investigate why.

Problem 2: Investigating Eigenvalue Computation

Consider the matrix

$$A = \begin{pmatrix} 10 & 9 & 0 \\ 0 & 10 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

You have two approaches for eigenvalue computation available from the course:

- Power iteration;
- QR iteration.

Design a computational experiment to investigate their behaviour.

Tasks

1. Use **power iteration** to approximate an eigenvalue of A . Investigate how the choice of the initial vector affects the computed result.
2. Implement **QR iteration** and use it to compute approximations to all eigenvalues of A .
3. For both methods, choose appropriate quantities to monitor convergence and produce suitable plots.
4. Compare the information obtained from the two approaches. In particular, investigate why power iteration does not provide the same information as QR iteration.
5. Now consider the perturbed matrix

$$A_\varepsilon = A + \varepsilon \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

for several values of ε .

Repeat your experiments and investigate how the eigenvalue computations change as ε varies.

6. Use `np.linalg.eig` only as an independent check of your results.
7. Write a short conclusion explaining what your experiments reveal about the strengths and limitations of power iteration and QR iteration.

Question to think about

Power iteration and QR iteration are both based on repeatedly applying orthogonalization-related ideas, but they extract spectral information in very different ways.

What does power iteration discover that QR iteration does not need to assume?

Use your numerical experiments to support your answer.

Problem 3: Construct a Difficult Linear System

Consider the problem of solving $Ax = b$, where A is a nonsingular matrix. Instead of being given a difficult matrix, your task is to *construct* one.

The purpose of this experiment is to investigate the difference between

[difficulty of the problem and difficulty of the algorithm.]

Tasks

1. Generate a nonsingular matrix A and choose a known vector x_{exact} .

2. Construct the right-hand side using

$$b = Ax_{\text{exact}}.$$

3. Solve the resulting system numerically using at least two numerical methods studied in the course.

4. Measure the relative solution error

$$E_x = \frac{\|x - x_{\text{exact}}\|}{\|x_{\text{exact}}\|},$$

and the relative residual

$$E_r = \frac{\|Ax - b\|}{\|b\|}.$$

5. Now deliberately modify the matrix so that the linear system becomes increasingly difficult to solve accurately.

6. Investigate at least three different ways of constructing difficult test matrices.

7. For each experiment, record quantities such as

$$\text{cond}(A), \quad E_x, \quad E_r.$$

8. Compare the behaviour of the different numerical methods.

9. Determine whether the observed loss of accuracy is primarily associated with

- the conditioning of the problem,
- the numerical algorithm, or
- floating-point arithmetic.

10. Construct at least one example for which

the residual is small but the solution error is large.

11. Explain carefully why this situation can occur.